COM205 Microelectronic Devices and Circuits

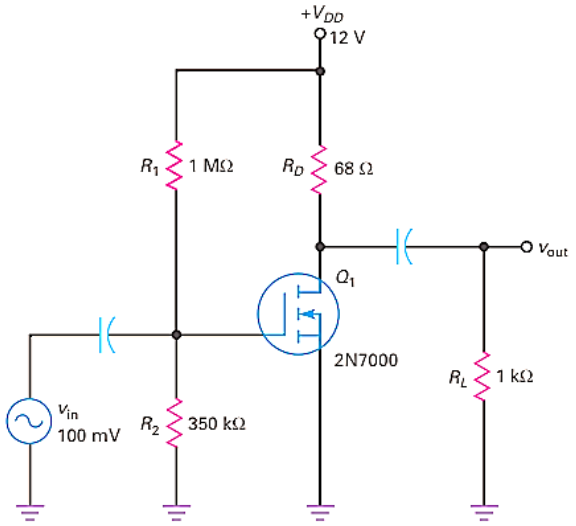
Quiz 2 Questions | Session 2025-04

Question 1 [15m]

(a) Explain bel, decibel, power gain and voltage gain. [0.5x4=2m]

(b) Analyze the effect of internal, bypass and coupling capacitors in frequency response of an amplifier [1x3=3m]

(c) For the EMOSFET circuit, find V_{GS}, I_D, g_m, and V_{out}. The MOSFET specifications are k = 104 x10^{-3} A/V^2, $I_{D(on)}$ = 600 mA, and $V_{GS(th)}$ =2.1 V. Assume $I_D = k [V_{GS} - V_{GS(th)}]^2$ [1.5x4=6m]



(d) For the filter below determine the cutoff frequencies and draw the frequency response curve. Assume that the capacitor C3 has very low impedance at midband frequencies. [3+1=4m]

